

SYNOPSIS

SuhanaGhar AI: Climate-Responsive Smart Housing System

STUDENT & INSTITUTIONAL INFORMATION

- **University:** Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal
- **Degree:** Bachelor of Technology (B.Tech)
- **Branch:** Computer Science Engineering - Artificial Intelligence & Machine Learning
- **Project Guide:** Ms. Akanksha Meharam, Department of Computer Science & Engineering

Name	Roll No.
Yogesh Pawar	0105AL231233
Yash Kumar Patle	0105AL231230
Vinay Patel	0105AL231227
Raj Patel	0205AL243D15

1. Project Title

SuhanaGhar AI: A Machine Learning Framework for Climate-Responsive Residential Design in India

2. Project Category

Minor Project / Academic Research Project

Domain: Artificial Intelligence & Machine Learning

Sub-Domain: Sustainable Technology / Green Building

3. Team Details

Name	Roll No.
Yogesh Pawar	0105AL231233
Yash Kumar Patle	0105AL231230
Vinay Patel	0105AL231227
Raj Patel	0205AL243D15

Department: Artificial Intelligence & Machine Learning

Institution: Oriental Institute of Science and Technology, Bhopal

4. Project Duration

Activity	Duration
Planning & Research	2 Weeks
Data Collection & Preprocessing	2 Weeks
Model Development & Training	3 Weeks
Frontend Development	3 Weeks
Testing & Documentation	2 Weeks

Activity	Duration
Total Duration	12 Weeks (3 Months)

Academic Year: 2025-2026

5. Introduction

The residential building sector accounts for approximately **30-40% of India's total electricity consumption**, with air conditioning being the primary contributor. Most Indian homes are designed without considering local climate conditions, solar heat gain, building orientation, or passive cooling principles. This results in:

- High electricity bills
- Poor indoor thermal comfort
- Increased carbon footprint
- Inefficient house planning

SuhanaGhar AI addresses these challenges by providing an **intelligent, data-driven platform** that predicts heating and cooling loads and provides smart architectural recommendations for energy-efficient home design.

6. Problem Statement

Conventional residential house designs in India ignore:

Ignored Factor	Consequence
Local climate conditions	Inappropriate design for the region
Solar heat gain dynamics	Excessive cooling needs
Building orientation	Higher energy consumption
Natural ventilation	Dependency on mechanical cooling
Passive cooling principles	Missed opportunities for free cooling

Result: Homes that are uncomfortable, expensive to operate, and environmentally damaging.

7. Objectives

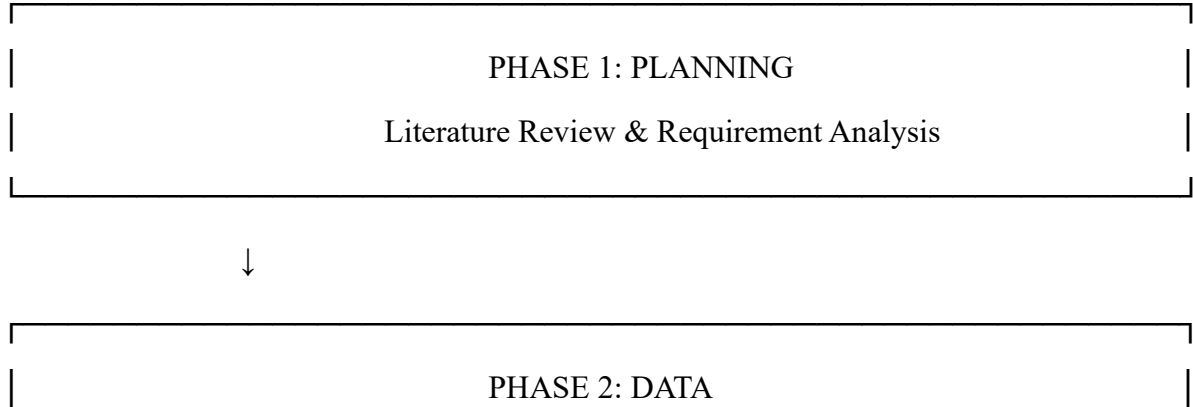
Primary Objectives

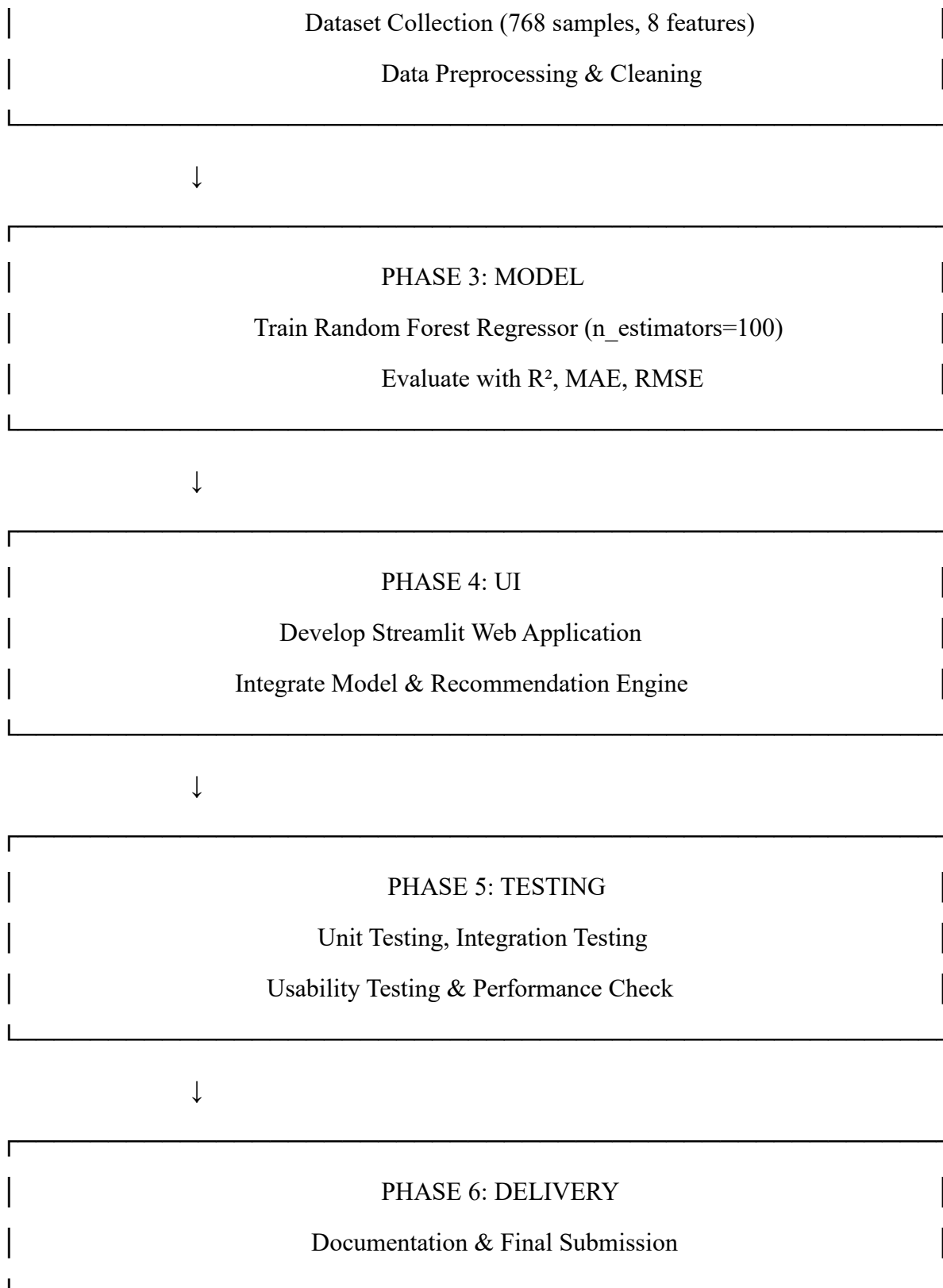
No.	Objective
1	Develop ML model to predict heating and cooling loads
2	Create web-based interface for easy access
3	Provide climate-responsive design recommendations
4	Educate users on energy efficiency parameters

Secondary Objectives

No.	Objective
1	Identify most significant energy-impacting parameters
2	Demonstrate AI application in sustainable architecture
3	Create scalable framework for future enhancements

8. Methodology





9. Technology Stack

Category	Technology	Version
Language	Python	3.9+
Frontend	Streamlit	1.28+
ML Framework	Scikit-learn	1.3+
Data Handling	Pandas, NumPy	2.0+, 1.24+
Visualization	Matplotlib, Plotly	3.7+, 5.14+
API	Requests	2.31+
Model Export	Pickle	-

10. Expected Outcomes

Outcome	Description
☒ ML Model	Random Forest model with >85% accuracy
☒ Web Application	Fully functional Streamlit dashboard
☒ Prediction System	Real-time heating & cooling load prediction
☒ Design Advisor	Climate-specific architectural recommendations
☒ Documentation	Complete project report and user manual

11. Timeline / Work Plan

Week	Activity	Deliverable
1-2	Literature Review & Requirements	Project Plan
3-4	Data Collection & Preprocessing	Clean Dataset
5-7	Model Training & Evaluation	Trained Model (model.pkl)
8-10	Streamlit App Development	Working Application
11	Testing & Bug Fixes	Test Report
12	Documentation & Submission	Final Report

12. Budget Estimate

Item	Cost (₹)
Software (Open Source)	0
Hardware (Existing System)	0
API Costs (Weather API - Free Tier)	0
Internet & Miscellaneous	500
Printing & Binding	1000
Total	₹1,500

13. References (Preliminary)

1. Tsanas, A., & Xifara, A. (2012). 'Energy performance of residential buildings using machine learning'. *Energy and Buildings*, 49, 560-567.
2. Breiman, L. (2001). 'Random Forests'. *Machine Learning*, 45(1), 5-32.

3. Givoni, B. (1998). *Climate Considerations in Building and Urban Design*. Van Nostrand Reinhold.
 4. UCI Machine Learning Repository. (2012). *Energy Efficiency Dataset*.
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14. Keywords

Machine Learning Random Forest Energy Efficiency Smart Housing Climate-Responsive Design Sustainable Architecture Heating Load Cooling Load Streamlit Green Building

15. Signature

Prepared by:

Yogesh Kumar
(Student)

Date: [Current Date]

Approved by:

Project Guide
(Department of AI & ML)

END OF SYNOPSIS